# System and Network Security

**Dr. Ashok Kumar Das**

**Professor**
Center for Security, Theory and Algorithmic Research
International Institute of Information Technology, Hyderabad

E-mail: *ashok.das@iiit.ac.in*
URL: `https://www.iiit.ac.in/faculty/ashok-kumar-das/`
`https://sites.google.com/view/iitkgpakdas/`

# Internet Security Protocol (IPSec) (Part 1)

# IPSec

- IPSec is a collection of protocols designed by the Internet Engineering Task Force (IETF) to provide security for a packet at the network layer.
- The network layer in the Internet is often referred to as the Internet Protocol or IP layer.
- IPSec helps to create authenticated and credential packets for the IP layer.

# IPSec

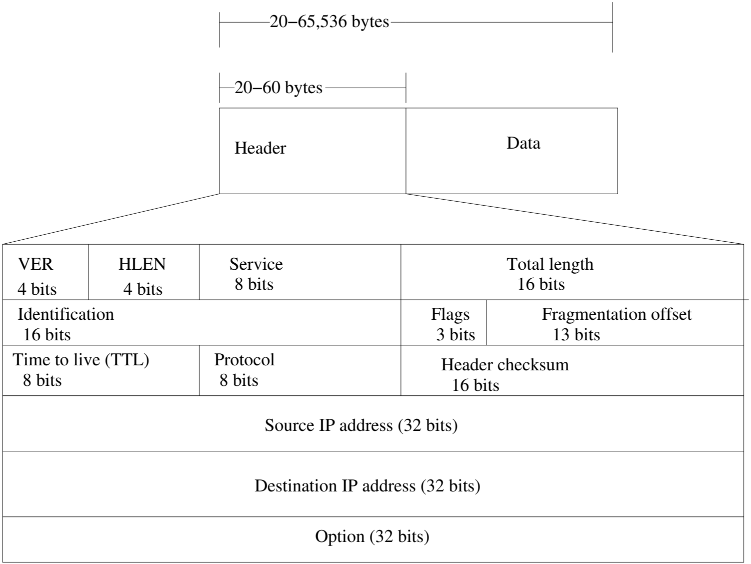
- IPSec is useful in several areas:
  - It can enhance the security of those client/server programs, such as electronic mail (e-mail), that uses their own security protocols.
  - It can enhance the security of those client/server programs, such as HTTP, that use the security services provided at the transport layer.
  - It can provide security for those client/server programs that do not use the security services provided at the transport layer.
  - It can provide security for node-to-node communication programs such as routing protocols.

# The Internet Protocol Version 4 (IPv4)

- IPv4 is the delivery mechanism used by the TCP/IP protocols.
- IPv4 is an unreliable datagram protocol - a "best-effort delivery" service, that is, it provides no error control or flow control (except for error detection on the header).
- IPv4 is a connection-less protocol for packet-switching network that uses the datagram approach.
- **Connectionless protocol:** A connectionless protocol is a communication method that sends data packets without establishing a connection between devices. It's similar to sending an unregistered letter in the post-office, where the recipient might not be notified if the letter doesn't arrive.
- **Connection-oriented protocol:** A connection-oriented protocol is a communication protocol that establishes a connection between two devices before data is exchanged. This connection is maintained for the duration of the communication.

# Datagram in IPv4

Packets in the IPv4 layer called datagrams.



# Datagram in IPv4

- Datagram is a variable-length packet consisting of two parts: 1) **header** and 2) **data**.
- The header is 20 to 60 bytes in length and contains information essential to routing and delivery.
- Brief description of IP header fields:
  - **Version (VER):** Defines the version of IP protocol. Currently the version is 4. However, version 6 (or IPng) may totally replace version 4 in the future.
  - **Header length (HLEN):** Defines the length of the datagram header in 4-bytes words. When there are NO options, the header length is 20 bytes, and the value of this field is 5 ($5 \times 4 = 20$). When the option field is at its maximum size, the value of this field is 15 ($15 \times 4 = 60$). Thus, $5 \leq$ HLEN $\leq 15$.
  - **Total length:** This 16-bit field defines the total length of the IPv4 header. To find the length of the data coming from the upper layer, we need the following calculation:
    length of data = total length - header length, where header length = HLEN $\times 5$. Max. datagram size = $2^{16} = 65,536$ bytes.

# Datagram in IPv4

- Brief description of IP header fields (continued...):
  - **Time to live (TTL):** A datagram has a limited lifetime in its travel through the Internet. This 8-bit field prevents a packet from traveling a loop. The sender sets a value, that is decremented at each hop. If it reaches zero, the packet is discarded.
  - **Protocol:** This 8-bit field defines the higher-level protocol that uses the services of the IPv4 layer.

Table: Protocol values

| Value | Protocol |
|---|---|
| 1 | ICMP (Internet Control Message Protocol) |
| 2 | IGMP (Internet Group Management Protocol) |
| 6 | TCP (Transmission Control Protocol) |
| 17 | UDP ( User Datagram Protocol) |
| ⋮ | ⋮ |

# Datagram in IPv4

- Brief description of IP header fields (continued...):
  - **Header checksum:** This 16-bits field in the IPv4 packet covers only the header, not the data.
    The checksum is computed as follows:
    - Header is treated as a sequence of 16-bit integer.
    - The integers are all added using 1's complement arithmetic.
    - 1's complement of the final sum is taken as the checksum.

    If there is any mismatch in checksum, then that will cause the datagram to be discarded.
  - **Source IP address:** This 32-bit field defines the IPv4 address of the source. This field must remain unchanged during the time the IPv4 datagram travels from the source host to the destination host.
  - **Destination IP address:** This 32-bit field defines the IPv4 address of the destination. This field must also remain unchanged during the time the IPv4 datagram travels from the source host to the destination host.

# Datagram in IPv4

- Brief description of IP header fields (continued...):
  - **Options:** The header of the IPv4 datagram is made of two parts: a fixed part and a variable part. The fixed part is 20 bytes long. The variable part comprises the options that can be a maximum of 40 bytes. The options are not required for a datagram. They can be used for network testing and debugging purposes.
  - **Identification:** This 16-bit field identifies a datagram originating from the source host. The combination of the identification and source IPv4 address must uniquely define a datagram as it leaves the source host.
  - **Flag:** This is a 3-bit field, in which the first bit is reserved, the second bit is called the "do not fragment (D)" bit, and the third bit the "more fragment (M)" bit.
    - If D bit is 1, the machine must not fragment the datagram.
    - If D bit is 0, the datagram can be fragmented if necessary.
    - If M bit is 1, its meaning is that the datagram is NOT the last fragment; there are more fragments after this one.
    - If M bit is 0, it means that this is the last or only segment.

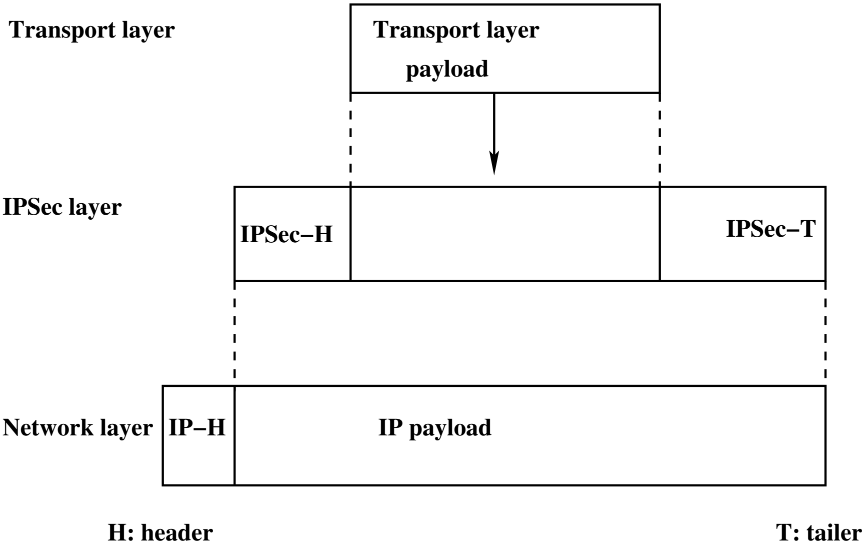
INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY HYDERABAD

# Different Modes

# Transport mode

- In this mode, IPSec protects what is delivered from the transport layer to the network layer.
- In other words, transport mode protects the network layer payload, the payload to be encapsulated in the network layer.
- Note that the tarnsport mode does not protect the IP header.

# IPSec in transport mode



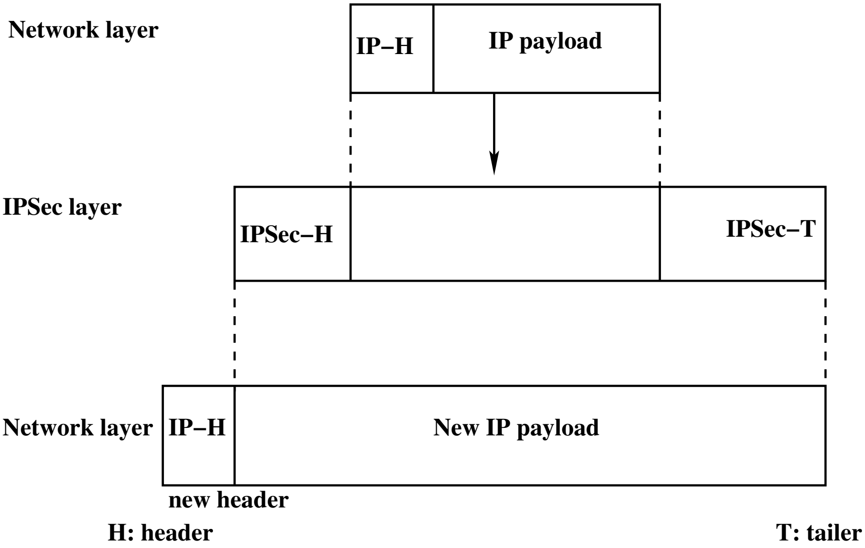
# Transport mode

- This mode is normally used when we need host-to-host (end-to-end) protection of data.
- The sending host uses IPSec to authenticate and/or encrypt the payload delivered from the transport layer.
- The receiving host uses IPSec to check the authentication and/or decrypt the IP packet and deliver it to the transport layer.

# Tunnel mode

- In this mode, IPSec protects the entire IP packet.
- It takes an IP packet, including the header, applies IPSec security methods to the entire packet, and then adds a new IP header.

# IPSec in tunnel mode

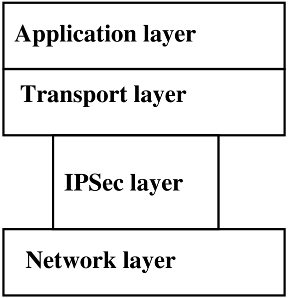


# Tunnel mode

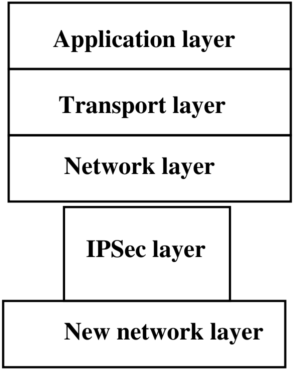
- The new IP header has different information than the original IP header.
- Tunnel mode is normally used between two routers, between a host and a router, or between a router and a host.
- In other words, tunnel mode is used when either the sender or the receiver is not a host.
- The entire original packet is protected from intrusion between the sender and the receiver, as if the whole packet goes through an imaginary tunnel.

# Transport mode *versus* Tunnel mode

- In transport mode, the IPSec layer comes between the transport layer and the network layer.
- In tunnel mode, the flow is from the network layer to the IPSec layer and then back to the network layer again.




(a) Transport mode




(b) Tunnel mode

# Two IPSec Security Protocols

- Authentication Header (AH)
- Encapsulating Security Payload (ESP)

**Purpose of these protocols to provide authentication and/or encryption for packets at the IP level.**

# Authentication Header (AH) Protocol

# Authentication Header (AH) Protocol

- Designed to authenticate the source host and to ensure the integrity of the payload carried in the IP packet.
- Uses a hash function and a symmetric key to create a message digest; the digest is inserted in the authentication header.
- The AH is then placed in the appropriate location based on the mode (transport or tunnel).
- Here we discuss the AH protocol under the transport mode.

# Authentication Header (AH) Protocol

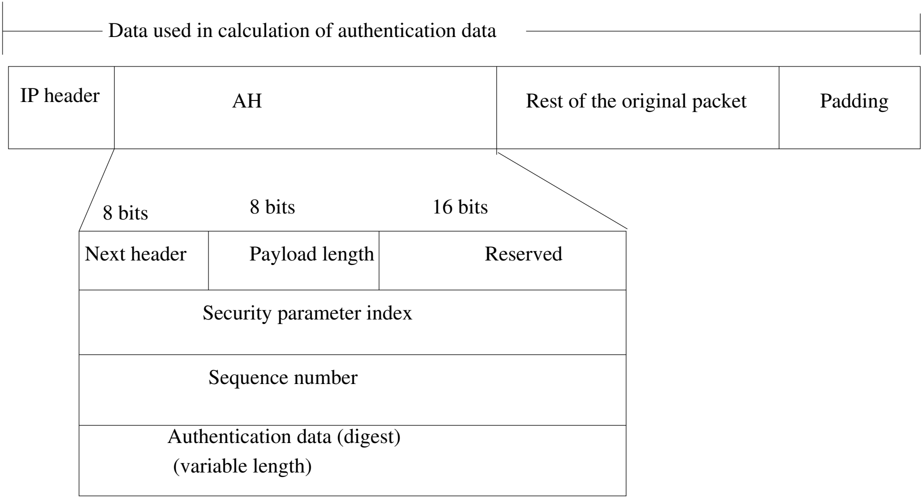



Figure: AH protocol

# Fields for Authentication Header (AH) Protocol

- **Next header:** The 8-bit next header field defines the type of payload carried by the IP datagram (such as TCP, UDP or ICMP).
- **Payload length:** This 8-bit field does not define the length of the payload; it defines the length of the authentication header in 4-byte multiples, but it does not include the first 8-bytes.
- **Sequence number:** A 32-bit sequence number provides ordering information for a sequence of datagrams. The sequence numbers prevent a playback/replay attack.
  - The sequence number is not repeated even if a packet is retransmitted.
  - A sequence number does not wrap around after it reaches $2^{32}$; a new connection must be established.
- **Authentication data:** The authentication data field is the result of applying a hash function to the entire IP datagram except for the fields that are changed during transmit (e.g., time-to-live (TTL)).

# Steps in Authentication Header (AH) Protocol

- **Step 1:** An authentication header is added to the payload with the authentication data field set to 0.
- **Step 2:** Padding may be added to make the total length even for a particular hashing algorithm.
- **Step 3:** Hashing is based on the total packet. However, only those fields of the IP header that do not change during transmission are included in the calculation of the message digest (authentication data).
- **Step 4:** The authentication data are inserted in the authentication header.
- **Step 5:** The IP header is added after changing the value of the protocol field to 51.